7.2.6 Symmetric and Skew-symmetric Matrices

Definition 7.17

A square matrix A is said to be **symmetric** if $A^T = A$.

That is, $A = [a_{ij}]_{n\times n}$ is a symmetric matrix, then $a_{ij} = a_{ji}$ for all i and j.

For instance, $A = \begin{bmatrix} 3 & -6 & 9 \\ -6 & 8 & 5 \\ 9 & 5 & 2 \end{bmatrix}$ is a symmetric matrix since $A^T = A$.

Observe that transpose of A^T is the matrix A itself. That is $\left(A^T\right)^T = A$.

Definition 7.18

A square matrix A is said to be **skew-symmetric** if $A^T = -A$.

If $A = [a_{ij}]_{n\times n}$ is a skew-symmetric matrix, then $a_{ij} = -a_{ji}$ for all i and j.

Now, if we put $i = j$, then $2a_{ii} = 0$ or $a_{ii} = 0$ for all i. This means that all the diagonal elements of a skew-symmetric matrix are zero.

For instance, $A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}$ is a skew-symmetric matrix since $A^T = -A$.

It is interesting to note that any square matrix can be written as the sum of symmetric and skew-symmetric matrices.

Theorem 7.1

For any square matrix A with real number entries, $A + A^T$ is a symmetric matrix and $A - A^T$ is a skew-symmetric matrix.

Proof

Let $B = A + A^T$.

$$B^T = (A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T = B.$$

This implies $A + A^T$ is a symmetric matrix.

Next, we let $C = A - A^T$. Then we see that

$$C^T = (A + (-A^T))^T = A^T + (-A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T) = -C$$

This implies $A - A^T$ is a skew-symmetric matrix.

Theorem 7.2

Any square matrix can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

Proof

Let A be a square matrix. Then, we can write

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

From Theorem 7.1, it follows that $(A + A^T)$ and $(A - A^T)$ are symmetric and skew-symmetric matrices respectively. Since $(kA)^T = kA^T$, it follows that $\frac{1}{2}(A + A^T)$ and $\frac{1}{2}(A - A^T)$ are symmetric and skew-symmetric matrices, respectively. Now, the desired result follows.

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